Symbols used in 'Einstein-Rosen bridges in the gravitational fields of dirac16complex'

symbol	meaning	definition
t, r, θ, ϕ	Schwarzschild coordinates	$ds^2 = -(1 - \frac{2m}{r})dt^2 + \frac{dr^2}{1-2m/r} + r^2 d\Omega^2$
$d\Omega^2$	sphere metric	$d\theta^2 + \sin^2\theta\, d\phi^2$
m	Schwarzschild mass	geometric units $G = c = 1$
u	Einstein-Rosen coordinate	$u^2 = r - 2m$ (neutral), $u^2 = r^2 - q^2$ (charged), $x_0 = x_h + su^2$ (Section 2)
q	charge parameter	$F_{tr} = q/r^2$
$F_{\mu\nu}$	electromagnetic field tensor	antisymmetric, only F_{tr} non-zero
$T^{EM}_{\mu\nu}$	electromagnetic stress tensor	$\frac{1}{4\pi}(F_{\mu\alpha}F_{\nu}{}^{\alpha} - \frac{1}{4}g_{\mu\nu}F^2)$
$g_{\mu\nu}$, g	metric and its determinant	
$R_{\mu\nu}$, R, $G_{\mu\nu}$	Ricci tensor, scalar, Einstein tensor	$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}$
$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$	Kretschmann scalar	full contraction of the Riemann tensor
x_0, x_4	hidden coordinate, time	coordinates of the dirac16complex project
f	metric function	modern $1 - r_s/x_0$; future $1 - r_s/x_0 - H_\Lambda^2 x_0^2$
r_s, H_Λ	mass and dark-energy parameters	numerics: $r_s = 1$, $H_\Lambda = 0.1$
x_h	horizon	simple root of f
s	sign	$+1$ if $f'(x_h) > 0$, -1 if $f'(x_h) < 0$
g_{uu}, g_{44}	metric components in (u, x_4)	$g_{uu} = 4u^2/f$, $g_{44} = -f$
κ	surface gravity	$\lvert f'(x_h)\rvert/2$
H	Hubble parameter of the primordial field	vielbein $\mathrm{diag}(\tan 6Hx_0, \dots)$
ρ	tortoise coordinate	$d\rho = dx_0/f$
U, V	Kruskal coordinates	$U = -e^{-\kappa(x_4-\rho)}$, $V = e^{\kappa(x_4+\rho)}$ on sheet I
ϵ	Killing energy of a universe (Kohn-Sham mode)	mode $\propto e^{-i\epsilon(x_4-\rho)}$
β_ϵ	Bogoliubov coefficient	$\lvert\beta_\epsilon\rvert^2 = 1/(e^{2\pi\epsilon/\kappa} + 1)$
T_H	Hawking temperature	$\kappa/(2\pi)$
H_I, H_{II}	Killing Hamiltonians of the two sheets	the bridge vacuum is annihilated by $H_I - H_{II}$